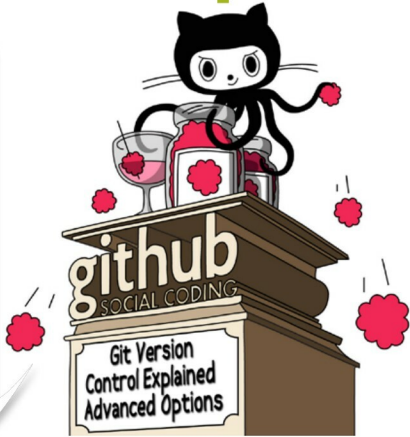


Git Version Control Explained Advanced Options

The article published in the September issue covered the fundamentals of Git, including the basic data structures and commands to get started with a local repository and work with a remote repository. This article will focus on the advanced features and commands available, using an example to demonstrate how to collaboratively integrate with GitHub for coding.



Let's take a look at the advanced features and commands available in Git.

Merging branches

In Git, you create short-lived branches to develop some feature. Once the feature is complete, you merge the code to the main branch, and delete the feature branch or topic-branch. To merge a branch to the current branch, the command `git merge branch_name` is used.

Pulling changes from a remote branch

When you want to sync changes from a remote repository, use the `git pull` command—`git pull branch_name`, for example; though, if you are pulling from the origin repo, just `git pull` is enough.

This will fetch changes from the remote repository and will try to merge them to the current local repository. Thus,

`git pull` is the same as `git fetch` or `git merge`. If multiple people are committing to a repository, `git pull` is not recommended—you should use `git fetch + rebase` (see the next section).

Rebasing changes

The `merge` command tries to put the commits from the branch to be merged on top of the `HEAD` of the current local branch checkout. The `git rebase branch_name` command is a type of merge; it tries to find out the common ancestor between the current local branch and the branch you are trying to merge. It then puts the commits from the branch you are trying to merge to the current local branch by modifying the order of commits in the current local branch. For example, if the current local branch has the commits A-B-C-D, and the merge branch has the commits A-B-X-Y, then the merge command will try to turn the local branch into something like A-B-C-D-X-Y. The rebase command, however, tries to formulate the repo as A-B-X-Y-C-D.